

Monitoring volatile homicide trends across U.S. cities*

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*Data and code to replicate the findings can be obtained from
https://www.dropbox.com/s/zp9zxjuq01xoxk1/Disseminate_HomicideTrends.zip?dl=0.

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Title: Monitoring Volatile Homicide Trends Across U.S. Cities

Abstract: While U.S. homicide rates remain at near historical lows, large year-to-year percentage increases in homicide in 2015 and expected for 2016 have generated much speculation about its causes among policymakers, academics, criminal justice practitioners, media representatives and other social commentators. In this article we show how two data visualization tools, funnel charts and time series fan charts, can show the typical volatility in homicide rates in different cities over time. Many of the recent increases are not out of the norm given historical patterns, and so one need not rely on various ex-ante hypotheses to explain homicide spikes occurring in some U.S. cities.

Keywords: Data-Visualization, Funnel-Charts, Fan-Charts, Homicide-Rates, Time-Series

Data and code to replicate the analysis can be downloaded from

https://www.dropbox.com/s/ln7i8qb2dwxuhno/Disseminate_HomicideTrends_Anon.zip?dl=0.

Introduction

In the aftermath of several incidents involving police officer shootings of minority citizens, the media has focused on increases in homicide rates in several cities including St. Louis, Chicago and Baltimore. These increases have been linked to particular events occurring within those cities such as civil unrest and a reduction of police activity (University of Chicago Crime Lab, 2017; Dewan, 2017; Fagan & Richman, 2017; Morgan & Pally, 2016; Rosenfeld, 2015; Towers & White, 2017). To explore the validity of such claims, this paper introduces two graphical methods, funnel charts and time series fan charts, to evaluate whether homicide rates are atypical between cities in any particular year or within cities over time. The application of these techniques allows us to advance the understanding of homicide trends and to offer researchers better tools for analyzing crime data both of which may serve to better inform the media, public and policymakers.

Here we show that homicide rates are a volatile measure, and prior attempts to identify extreme changes over-time, in particular percent changes in homicides, produce exaggerated evidence of change. Homicides are rare events, thus rates in cities with smaller populations will tend to have more volatile fluctuations over time. Using funnel charts, we will show that several cities, such as Chicago, Baltimore, and St. Louis, have had historically high homicide rates compared to the cities of similar population sizes consistently since at least the 1980's. Generating prediction intervals using a random effects binomial regression model will also show that recent increases in those cities are not atypical given historical homicide rates over time.

Identifying Changes in Homicide Rates over Time

One of the most common ways to identify recent increases in homicides over time is to calculate a percent change. This approach is common both in the popular press¹, as well as in academic articles (Baumer, Wolff, 2014; Parker, Mancik & Stansfield, 2016; Rosenfeld, 2016; Stamatel, 2014). For example, given a homicide rate of 10 (per 100,000) in the year 2000, and a homicide rate of 12 in the year 2010, the percent change over the ten year period would be $(12-10)/10 = 0.2$, or 20%. Although percent change is simple to calculate and interpret, it has particularly poor properties when examining low rate homicide trends. First, the variance of the metric is not defined (Wheeler, 2016), so an actual outlying value is difficult to identify. Second, the variance of the metric is influenced by the denominator, so places with lower homicide rates will have more volatile changes from year to year. Third, the variance of the metric is asymmetric; a change of a homicide rate from 4 to 5 is a 25% increase, whereas a change from 5

¹ Some recent examples describing the homicide increase in terms of percent change can be seen at the *New York Times* (Davey and Smith, 2015), the *Washington Post* (Berman, 2016), the *Economist* (Economist, 2017), and 538 (Ahser, 2016; Bialik, 2015).

to 4 is only a 20% decrease. This subsequently will result in larger percentage increases than percentage decreases.

Highly varying percent changes can be seen in reported samples of homicide rates. Stamatel (2014) shows decreases of 406% to increases of 100% for homicide rates in various countries from 1956 through 1998. Baumer and Wolff (2014) show percent decreases in large U.S. cities of up to 80% from 1993 to 2001. Parker, Mancik & Stansfield (2016) show percent changes of -90% to +200% in a sample of 199 U.S. cities for more recent crime trends from 2007 through 2011. Rosenfeld (2016) shows percent changes in a selected sample of large U.S. cities of +120% to -40% for changes in homicides from 2014 to 2015. Such ranges suggest that even percent changes of over 100% for large polities are not uncommon and should not per-se be considered unexpected simply by chance. Because of this year-to-year variability researchers often aggregate homicide rates over multiple year spans and analyze those, see Beaulieu and Messner (2009) for one example.

A more formal, statistical approach researchers have used to identify changes in homicide trends are to estimate structural breaks in trends over time (Cook & Cook, 2011; Messner et al., 2005; Piehl et al., 2003; Pyrooz et al., 2016). For example, Pyrooz et al. (2016) in trying to identify a Ferguson effect², use random effect growth models and test for linear change in the pre to post crime rates in 81 cities exceeding a population of 200,000. Using 12 months of pre and post data, they identify several cities with large increases in homicide rates.

The approach in Pyrooz et al. (2016) relies on specifying the exact intervention date where the break would be associated, and also uses a restrictive functional form of homicide rates over time (a linear trend in the twelve months before and after). Messner et al. (2005) use

² The Ferguson hypothesis is that police were less proactive in response to the public backlash over the police shooting of an unarmed African-American male in Ferguson Missouri (Pyrooz, et al., 2016).

spline regression techniques, where the knots of the splines are not specified but estimated (using a model selection technique, step-wise regression) to identify the location of change points in annual homicide rates in cities of over 250,000 between 1979 and 2001. They related the different timing of structural breaks to epidemics, and suggested these breaks were largely identifying booms and busts for homicides related to the crack epidemic.

These techniques are limited in their ability to identify whether *recent* increases in homicides are atypical. First, to test for a structural break one needs multiple data points before and after the break (Piehl et al., 2003). One cannot identify whether one observation represents a structural break. Second, none of the prior research listed used models that take into account how homicide rates for smaller cities are more variable. While the majority of cited research uses selection criteria of only examining cities of over 100,000 or more population, we will show there is still a large amount of expected variation between cities of less than a million population as compared to higher population cities. That is, one would expect much larger year-to-year changes in the homicide rate for a city with a population of say 100,000 versus a city with a population of 1 million.

Data Sources

For this analysis we use the FBI UCR reported homicides and population estimates between 1960 and 2015 at the *agency* level. The UCR data from 1960 through 2015 were obtained from Interuniversity Consortium for Political and Social Research (ICPSR) UCR data series. As of the analysis, the full 2016 UCR data has not been published by the FBI, only the half-year 2016 estimates.

Although errors in UCR crime reporting data are well documented (Loftin & McDowall, 2010; Maltz & Targonski, 2002), they are sufficient for the present purposes. Nonetheless, we

take several steps to alleviate potential errors. First, we only include cities and years that have reported for the full twelve months. We do not rely on imputed information, and the models presented in the paper do not rely on contiguous years of reporting. Second, the use of the graphs in the paper easily identify erroneous *high* data entries or outliers (Maltz, 2010). Although we have undoubtedly missed some errors, we have provided an appendix that describes all of the data anomalies spotted and corrected in the historical agency level UCR data provided by ICPSR.

To identify outliers, two different data analysis and visualization techniques are shown. The first will use funnel charts to identify outliers between cities in any one particular year, and the second will use forecasts and fan charts to identify outliers in changes over time.

Funnel Charts for Between City Comparisons

It is inappropriate to compare homicide *counts* between different cities that have very different population sizes, therefore we calculate homicide rates per capita, typically in terms of per 100,000 population. Less appreciated by researchers is that there are different underlying variabilities in homicide rates between cities. For example, if the national homicide rate is 5 per 100,000, and the observed rates in different cities are simply due to sampling error, cities of a population size of 100,000 would be expected to have a homicide rate within 1 and 10 per 100,000 95% of the time.³ Whereas a cities with a population size of 1 million would be expected to have a homicide rate within 3.7 to 6.4 per 100,000 95% of the time. Thus, in this example, a homicide rate of 9 for the city of 1 million is a larger outlier than a homicide rate of 9 for a city of 100,000 relative to the national average.

³ This is based on the 0.025 and the 0.975 quantiles of the binomial distribution with a rate of 5/100,000 and repeated trials of 100,000 draws and 1,000,000 draws respectively. This is calculated in R via ``qbinom(c(0.025,0.975), size=100000, prob=rate)`` for the city of 100,000 and ``qbinom(c(0.025,0.975), size=1000000, prob=rate)/10`` for the city of 1 million where ``rate = 5/100000``.

Wainer (2007) describes several examples where parties made erroneous conclusions based on identifying outliers in data with varying errors around the measurements. For example, the push to create smaller schools was based on the observation that the majority of schools with high average standardized test scores had smaller enrollments. However, this is simply an artifact because smaller schools had more sampling error about the mean, and so many small schools also had very low standardized test scores.

Another example with similarities to homicide rates would be if one mapped areas with high or low rare cancer rates, one would mostly identify counties in rural areas for both (Gelman & Price, 1999). This again is because the rate is more volatile for places with fewer individuals, and even in large populations the standard error around those rates is quite large because cancer is a rare event.

A chart to display this information in a more succinct fashion is a funnel chart (Spiegelhalter, 2005). Academics may be most familiar with funnel charts from meta-analyses, in which one plots the estimated effect size of the study against the size of the sample (or some other measure of standard error). When considering homicide rates though, the standard error is directly a function of the population size for a city. In particular, if one has an estimate of the centralized value (such as the national homicide rate), one then estimates a measure of standard error around that proportion. Here we use the exact binomial Clopper-Pearson confidence interval to denote standard errors on the chart. One way to write the interval is (Agresti, 2002):

$$B(a/2; x, n - x + 1) < \pi < B(1 - a/2; x + 1, n - x)$$

Where B is the beta distribution at quantiles $a/2$ and $1 - a/2$, respectively. For a 95% confidence interval, one would estimate the 2.5th and the 97.5th quantile. The terms x and n refer to the first and second shape parameters of the beta distribution, respectively. The n term

represents the size of the city, and the x term is the expected number of homicides given the overall homicide rate, π , which is simply $\pi \cdot n$. The a term is set by the researcher, and the smaller it is set the larger the interval will be. Thus, a 99% interval will have a greater length than a 95% interval.

Cities with smaller populations are expected to have greater differences in homicide rates compared to cities with larger populations, and one draws these confidence limits as lines on a chart to denote when cities are outside of the expected bounds. Figure 1 displays a funnel chart for homicide rates for 3,279 agencies of over 10,000 population in 2015. The grey error bands signify a 99.9% confidence interval around the overall homicide rate within the sample of 4.9 per 100,000.

[INSERT FIGURE 1 HERE]

Figure 1 displays several cities outside of the funnel chart ranges, but the majority of locations follow the spray pattern according to their population. It also demonstrates that many, but not all, larger cities are outside of the error bar ranges. The largest city in the sample, New York City, with nearly 10 million residents, falls within the narrow band for the overall homicide rate for the entire nation. In general, larger cities have homicide rates significantly higher than the national average.

The graph also highlights small agencies that have exceptionally high reported homicide rates, although the expanding intervals do a good job of capturing the majority of small populations in the graph. East St. Louis (Illinois), Chester (Pennsylvania), Gary (Indiana), and Prichard (Alabama) stand out among the smaller cities.

Finally, the graph also highlights a grouped set of outlier cities that have very high homicide rates given their populations of over 100,000 but less than 1 million. This set of cities

includes St. Louis and Baltimore at the top, with Flint, Detroit, New Orleans, Birmingham, Kansas City, and Jackson slightly below.

Evidence *against* idiosyncratic hypotheses about recent homicide increases in those particular cities can be assessed by viewing such funnel charts over the past six decades. Figure 2 displays the same 99.9% confidence intervals for each decade from 1960 through 2010, and highlights many of the same cities as being exceptionally high over the historical time period for which we have measurements. For example, St. Louis has had high homicide rates compared to cities of similar sizes since the 1970's. The same is true for Baltimore since the 1980's. Any reasonable explanation for the current homicide trends in any of these particular cities must to take into account *the historically very high homicide rates* in those places, dating back to at least the 1980s.

Additionally, the funnel charts display the homicide increase and decrease over the five decades, with very few cities being clearly above the funnel in 1960, whereas more and more cities accumulate into the high homicide ranges from 1970 through 1990. The year 2000 then displays fewer outlying cities in the 10,000 to 100,000 population range. In 2010 one will recognize that several of the smaller cities, East St. Louis and Chester, have consistently had high homicide rates relative to other cities of similar size going back over 5 years. In the case of Gary, Indiana, one can track the decline of population over time as well, as it was a high outlier with over 100,000 population in 1980, but has declined to under 100,000 population (as of 2006) while maintaining a relatively high homicide rate.

[INSERT FIGURE 2 HERE]

While examining funnel charts over time provides one way to assess change, it is difficult to see the natural progression of cities over time, and it is not easy to assess whether *recent*

increases should be seen as outliers or signify a meaningful change in homicide rates. We address this in the next section by predicting future homicide rates and displaying fan charts.

Time Series Fan Charts for Within City Comparisons

The prior funnel charts are only sufficient to examine variation *between* cities. Such that one can identify if a particular city has a high homicide rate while taking into account the variability between the different population sizes between two cities. But, one is often interested if changes *within* one city over time are sufficient to identify whether homicide rates are substantively increasing. To identify those outliers, one needs a model of what would be expected in the future given historical homicide rates in a particular city.

To accomplish this, we fit a binomial random effects model⁴:

$$\log\left(\frac{\hat{p}_{it}}{1 - \hat{p}_{it}}\right) = \beta_1 + \lambda_i(\text{City}_i) + \gamma_t(\text{Year}_t) + \phi_{it}$$

This model estimates the homicide rate in City i and Year t as a probability $\hat{p}$. The estimated probability is a function of a random effect for each city, λ_i , a random effect for each year, γ_t , and an additional term, ϕ_{it} , a random effect for every unique year and city. The equation can be interpreted as follows; each city will have an intrinsic estimate for the overall homicide rate over all years in the sample. Each year will also have an estimated random effect, so for example the estimate for 1970 will differ from 2000 *for all cities*. Finally, the last term is a random effect for each unique city and year. This is included to model overdispersion in homicide rates after taking into account city and year fixed effects (Agresti, 2002; Spiegelhalter, 2005). The first term, β_1 , is the global intercept. The distribution of the random effects are forced to have a mean of zero, so this provides identification for the global intercept. We fit this model for cities that had over

⁴ The specific function to fit the model in R is ``glmer(cbind(Homicide, Population - Homicide) ~ (1|Year) + (1|ORI) + (1|Row), binomial, data=HomicideData)`` in the `lme4` package (Bates, Maechler, Bolker, & Walker, 2015). The posterior draws are subsequently estimated using the ``sim`` function in the `arm` package (Gelman & Su, 2016).

100,000 population at any point between 1960 through 2014. This results in a total of 562 cities and 27,208 city-years of homicide data.

This model is equivalent to a logistic regression, where each row is an individual in the population, and each is assigned 1 if they were murdered and 0 otherwise (Britt, Rocque, & Zimmerman, 2017). But the model can be fit with aggregated data, so there is no need to expand the dataset to many millions of individual cases. While interpreting homicide rates as a probability may seem strange, in terms of providing future predictions it is not logically any different than modelling the logarithm of the homicide rate per 100,000 population (McDowall, 2002). The homicide rate per 100,000 is simply one way to express a percent. In our experiments, this model provided the best coverage of future predictions, compared to negative binomial regression models and/or using fixed effects instead of random effects.

While this on its face appears substantially different than the more typical Poisson regression of homicide counts and using the population count as an offset (Osgood, 2000) it is not. This is because the Poisson distribution is a special case of the binomial distribution when the number of trials is very large (Kingman, 2005), which is the case for cities with populations over 100,000. But, the binomial regression model has the advantage that the intrinsic variance due to differing population counts is taken into account for future predictions.

While the model can undoubtedly be improved, we take this as a simplified baseline model, not unlike fixed effects models used in prior research (McDowell & Loftin, 2009; McDowall & Curtis, 2014). The random effects model has the added benefit over the fixed effects model of shrinking future predictions. The idea behind shrinkage is synonymous with regression to the mean. If a location has a homicide rate of 20 one year, and the national level homicide rate is 5, the best prediction going forward into the future is somewhere between 5 and

20 (Efron & Morris, 1977). These shrunk estimates tend to provide better forecasts than fixed effect estimates (Savitz & Raudenbush, 2009).

Next, we generate future *prediction intervals* by simulating 1,000 posterior draws from the random effects model (Gelman & Hill, 2007). Prediction intervals are substantially easier to interpret than confidence intervals. For example, given a 90% prediction interval, if correct that interval should cover future observations 90% of the time.⁵ Here we fit the model with data from 1960 through 2014, 2015 homicide rates are not used at all in the fitting process. Then we check coverage for the model by comparing posterior simulations for the prior year to the future year, e.g. posterior simulations for Chicago in 1970 are used to determine whether they cover the homicide rate in Chicago in 1971. We generate 50% prediction intervals based on the 25th and 75th quantiles of the posterior simulations, as well as 80% prediction intervals based on the 10th and 90th percentiles. These are displayed as varying shades in the graphs, using darker grey for the 50% interval and lighter grey for the 80% interval (Maltz & Zawitz, 1998). These are sometimes called fan charts (Spiegelhalter, Pearson, & Short, 2011).

Figure 3 displays an example fan chart for the city of New Orleans based on the random effects binomial model. The forecasted intervals are in grey, and the actual observed homicide rates are displayed as black circles. In the historical data for New Orleans, year to year changes of 10 to 20 in the homicide rate (up or down) are not uncommon. There is also a large spike in 2007 (the year in which Hurricane Katrina occurred), up to a homicide rate of over 90, and

⁵ The intercept for the actual estimated model is -9.962, and the standard deviation for the City, Year, and row random effects are 0.83, 0.27 and 0.34 respectively.

subsequent regression to more usual homicide rates hovering around 50. To be outlying based on this model, New Orleans would have needed a homicide rate of over 70 per 100,000 in 2015.⁶

[INSERT FIGURE 3 HERE]

To evaluate the overall effectiveness of the models' prediction intervals, we calculate how well the 50% and 80% prediction intervals cover the observed data. Table 1 displays those coverages for the historical (in sample data) and the out-of-sample 2015 data. For both, we can see that the model is not perfect, but has close to the expected coverage.

To read the table, the lower 25th percentile from the posterior simulations should have around 25% of the observed values lower than predicted.⁷ We see that in the historical sample lower values than predicted occur 30% of the time. So the lower bound appears to be slightly too high across the samples. We also see that the high interval has slightly too few outliers, at 20%. Examining the wider 80% interval, the same pattern occurs, and the lower interval has 19% below the predictions, but the upper interval is very close to nominal coverage with only 9% above the 90th percentile.

[INSERT TABLE 1 HERE]

In the out-of-sample 2015 data, we see that the coverage is very near to what is expected. Twenty five percent of the cases are above the 75th percentile, and ten percent of the cases are above the 90th predicted percentile. Again, the lower bounds are not low enough, but as most interest is in identifying cities with *rising* homicide rates the model appears to be effective.

⁶ While the charts here show homicide rates on a linear scale, one might also consider a logarithmic scale on the Y axis (Schmid, 1986). The semi-log time series chart is better to show proportional changes over time. Given homicide rates in these charts do not vary greatly they tend to look very similar to charts with a logarithmic scale.

⁷ In most circumstances, prediction intervals on out of sample data are *too small* (Hyndman & Athanasopoulos, 2014). So prediction intervals tend to cover actual observations at a smaller rate than expected.

Tables 2a and 2b display the agencies that have higher ($n = 53$) than expected homicide rates, and tables 3a, 3b, and 3c display those cities who have lower ($n=104$) than expected homicide rates in 2015. Each are based on the 80% prediction interval. The majority of the cities with high homicide rates are smaller and have around 100,000 population, with the exception of Milwaukee⁸ (population 600,400) and Nashville (population 658,029). There are several instances of state and county police agencies as well, such as the Connecticut State Police and St. Louis County Police. The tables are arranged so that the cities with the largest difference from predicted are sorted to the top. In Table 2a Portsmouth, VA was the largest high outlier, with an observed homicide rate of 28 per 100,000, which is far outside the 80% prediction interval of 7.5 to 13.7. For the low outlier cities the majority are cities that experience zero homicides. The largest low outlier is Lowell, Massachusetts (the first row of Table 3a), which experienced zero homicides, but had a lower prediction interval of 2 homicides per 100,000.

[INSERT TABLE 2a, 2b AND TABLE 3a, 3b and 3c HERE]

Figure 4 displays a set of panel charts showing several additional cities and their forecasted intervals. Dallas and New York City⁹ show the homicide decline, and given very little volatility over the prior years have subsequently smaller prediction intervals. In contrast to this, Chicago has shown larger fluctuations since 1960, and subsequently has larger prediction intervals. Based simply on the prediction interval for 2015, Chicago's homicide rate in 2016, 28 per 100,000 (University of Chicago Crime Lab, 2017) would not be outside, but near the upper limit of the 2015 80% prediction interval. Baltimore shows the widest prediction intervals

⁸ Similar to Baltimore, Milwaukee has come under national scrutiny due to cases of police use of force and subsequent civil unrest (Desmond, Papachristos, & Kirk, 2016).

⁹ New York City homicide rates do not include the homicides occurring on 9/11. The final homicide count of 649 for 2001 was taken from http://www.nyc.gov/html/nypd/downloads/pdf/analysis_and_planning/seven_major_felony_offenses_2000_2015.pdf.

compared to the other three cities, partly because it is a smaller city, but also because it bucks the trend of the majority of other cities (it has not shown similar homicide declines as compared to the rest of the dataset during the 1990's). The sharp rise in 2015 to a homicide rate of over 50 is outside the 75th percentile, but easily within the 90th percentile forecast.

[INSERT FIGURE 4 AND FIGURE 5 HERE]

Figure 5 displays fan charts for four cities that showed outlying values in 2015. Portsmouth, VA is quite high, but it appears in this case the model does a poor job of estimating the historical volatility in that city, as similar jumps were observed in the early 1990's. Milwaukee's homicide rate of 24 is only just above the predicted 90th percentile of 23. Lincoln and Boise demonstrate example cities that have lower homicide rates than predicted. This is due to examples of zero homicides when at least one would be expected. The upper intervals though appear to be reasonable. In 2016, Lincoln reported a total of 11 homicides, which results in a homicide rate of 4 per 100,000. This is just at the upper end of the predicted homicide rate in 2015.

Discussion

In this article we introduce two ways to evaluate a city's homicide rate. First, we introduce funnel charts to make more accurate comparisons between cities that have differing sizes of the population, and we introduce fan charts to show when cities have outlying values based on prior homicide rates in a particular city. With the funnel charts, we show that several cities with under 1 million population have had very high homicide rates relative to the national average for 30 years or more. Second, we illustrate the use of fan charts with prediction intervals based on a random effects binomial regression models. The prediction intervals for our models have appropriate coverage both with historical in-sample data, as well as out-of-sample 2015

data. Fan charts can be effectively used to identify outliers for a particular city based on past homicide rates.

While homicide rates remain substantially lower than they were at the peak of the crack epidemic of the late 1980s and early 1990s, the 10.8 percent increase in the national homicide rate between 2014 and 2015 (and the expected 8 percent increase for 2016) represents some of the largest year-to-year percentage increases seen in homicide rates in decades. In response, a number of hypotheses have been advanced to explain these seemingly sudden increases. Two of the most popular are civil unrest as a result of prominent cases of police shootings (Desmond, Papachristos, & Kirk, 2016; Morgan & Pally, 2016), and de-policing in reaction to criminal and civil litigation after incidents of police brutality, a.k.a the Ferguson effect (Fagan & Richman, 2017; Morgan & Pally, 2016; Pyrooz et al., 2016; Towers & White, 2017; Wolfe & Nix, 2016). But, as we show in this article, many of the cities exhibiting increases in homicide rates are often within future predicted intervals, and have had historically high crime rates compared to national averages. One need no special interpretation to many of the recent increases in homicide rates in cities besides simply knowing that such year-to-year changes are not unprecedented.

Many of the potential explanations popular in the media appear to be cherry picked to fit a popular narrative. For example, while decreases in arrests and stops are frequently cited to explain the rises in homicides in Chicago and Baltimore, New York City has seen similar decreases in stops due to civil litigation since 2011 (Sweeten, 2015), but has not seen subsequent increases in homicides in that time frame. Disseminating information about the expected variance of homicide rates over time in fan charts is a simple way to prevent the media, policymakers and the public from over interpreting by-chance fluctuations.

One can draw comparisons to the crime decline in the 1990's as well. There has been a plethora of theories about the mechanisms that resulted in the crime decline (Blumstein & Wallace, 2000; Levitt, 2004; Rosenfeld, Fornango, & Baumer, 2005). Despite accumulating evidence over time, there is relatively little consensus over what caused this decline (Farrell, Tilley & Tesloni, 2014). Large scale trends are based on observational data, so such evaluations tend to have low internal validity (Berk, 2005; Bushway & McDowall, 2006). Much of the narrative about the homicide increase follows this same rhetoric, albeit with a much smaller temporal sample. The graphs presented here should effectively illustrate the typical volatility in homicide rates, and prevent premature conclusions about whether some external shock is causing homicides to increase in any particular city.

What implications do our findings have for other researchers conducting analysis of homicide trends over time? Our research highlights two important insights. One, by identifying cities with very high levels of homicides relative to other cities of similar population size going back to the 1980's, the analysis demonstrates that whatever factors are responsible for cross-sectional differences in homicide rates between cities were established in the distant past, have remained relatively stable, and have continued to manifest themselves over the study period examined here. This suggests that long term-structural factors, such as residential segregation or concentrated disadvantage, may better explain homicide rates between cities over long periods of time (Pyrooz et al., 2016). Researchers focusing only on very recent homicide trends are likely to overestimate the effect of recent shocks compared to these factors. Second, we have shown that many cities have followed similar homicide trends over a long period of time. This suggests that national level explanations are important for understanding changes in homicide trends over time (McDowall & Loftin, 2009). More importantly, scholars attempting to explain idiosyncratic local

factors for homicide changes, such as de-policing or civil unrest, need to establish that the city under study is truly unique in its increase or decrease in homicide trends. Many of the recent explanations for increases in homicide in specific cities do not stand up to scrutiny when examining trends in cities that have experienced similar exogenous shocks, but have not witnessed the same increase in homicides.

As with all social science research, there are limitations to our study. First, funnel charts and fan graphs as presented do not incorporate any other demographic characteristics into the models. For funnel charts, these might include adjusting the homicide rate estimates per well established covariates of crime, such as poverty rates. However these covariates and their effects tend to be fairly temporally stable over time (Griffiths, 2013; Tcherni, 2011). As such, we would likely see similar patterns in the funnel charts over time. These covariates can also be incorporated into the time series fan charts; however, it is an open question as to whether they would reduce the size of the prediction intervals. That is a question for future researchers to examine.

Second, it is possible that the model could be improved by incorporating temporal aspects, such as an autoregressive term for the prior year's crime or a non-linear temporal trend component. But, if homicide rates are a random walk (McDowall, 2002), the best point estimate one can use to predict future rates is last year's homicide rate. While our model is simplistic, we provide good evidence that the model is adequate for identifying outlying high values. We show this via the models having appropriate coverage both in sample and out of sample. In regards to the lower bound, examining tables 3a to 3c show many cities with zero homicides (Loftin & McDowall, 2003), and zero-inflated models may improve the lower bound estimate.

Lastly, the reliance on the UCR data, which has a long history of problems with data reporting, is also a limitation of our study (Loftin & McDowall, 2010; Maltz, 1977; Maltz & Targonski, 2002). While the use of funnel and fan charts has the ability to spot erroneous high data entries, smaller cities regularly have zero homicides. If an agency had not reported and was mistakenly recorded as having zero homicides, such graphs are not as likely to uncover those data entry errors. Still, the same idea of funnel or fan charts can be applied to different sources of homicide data, such as at the county-level via the Vital Statistics reporting program, and applied the same way.

Finally, what is an appropriate policy response to these models? If Baltimore and Chicago show large homicide increases, but are within future prediction intervals, should there be a response by these cities? And if a city is outside of their prediction bands should policymakers or criminal justice professionals "do something" in response? The fact that homicides are rare and volatile make it challenging for researchers to identify whether the ups and downs are directly attributable to any particular exogenous factor. One should be cognizant of this limitation when attributing increases or decreases in homicide rates to *any* recent phenomena, as that logic is particularly susceptible to *post hoc ergo propter hoc*. To identify if a recent homicide increase is tied to a recent factor, say perceptions of police legitimacy, one would need to determine whether historical measures of legitimacy explain prior homicide rates both within a city and between cities over time. In short, one needs good social science research to ascertain causes of any homicide increase.

Understanding causality is important before any reasonable policy response can take place. For example, increases and decreases in homicide rates in Chicago have been largely attributed to public interpersonal disputes. Domestic homicides in Chicago have steadily

decreased since the 1960's (Block & Christakos, 1995) and more recent examination of these trends show that this continues to be the case (Wheeler & Block, 2017). Such patterns may not hold for all cities. If recent increases are due to domestic homicides, a much different policy response is required than if gun or gang violence are major contributors. Speculatively attributing any recent change in homicides to one factor and formulating a policy response is not likely to reduce future homicides.

Once a particular factor is identified, one can often draw on a wider array of evidence than homicide rates. For example, if gun violence is a major contributor to a recent increase in homicide, the researcher would examine whether other crimes involving guns, such as robbery and assault also increased in the same time period. Given these crimes are more prevalent than homicides, the data may provide clearer evidence as to whether local factors related to firearms also contributed to a recent increase in homicides.

Evidence arguing against this approach though is the fact that assaults and homicides may differ in general trends (McDowall & Curtis, 2015). It is also the case that homicides are often examined because they are viewed as more reliable measures of violent crime due to differential reporting (Loftin & McDowall, 2010). Some speculated factors that influence crime trends, such as perceptions of police legitimacy, may also influence reporting of victimization to the police (Desmond, Papachristos, & Kirk, 2016). Thus making it necessary to rely on homicide statistics that are not potentially biased due to changes in reporting patterns or discovery by the police.

Subsequently if one does want to make specific statements about increasing or decreasing homicide rates, it is essential to understand the typical uncertainty in those rates for cities of

different sizes, and the volatility of homicide rates over time. The graphs displayed in this article effectively accomplish both those goals.

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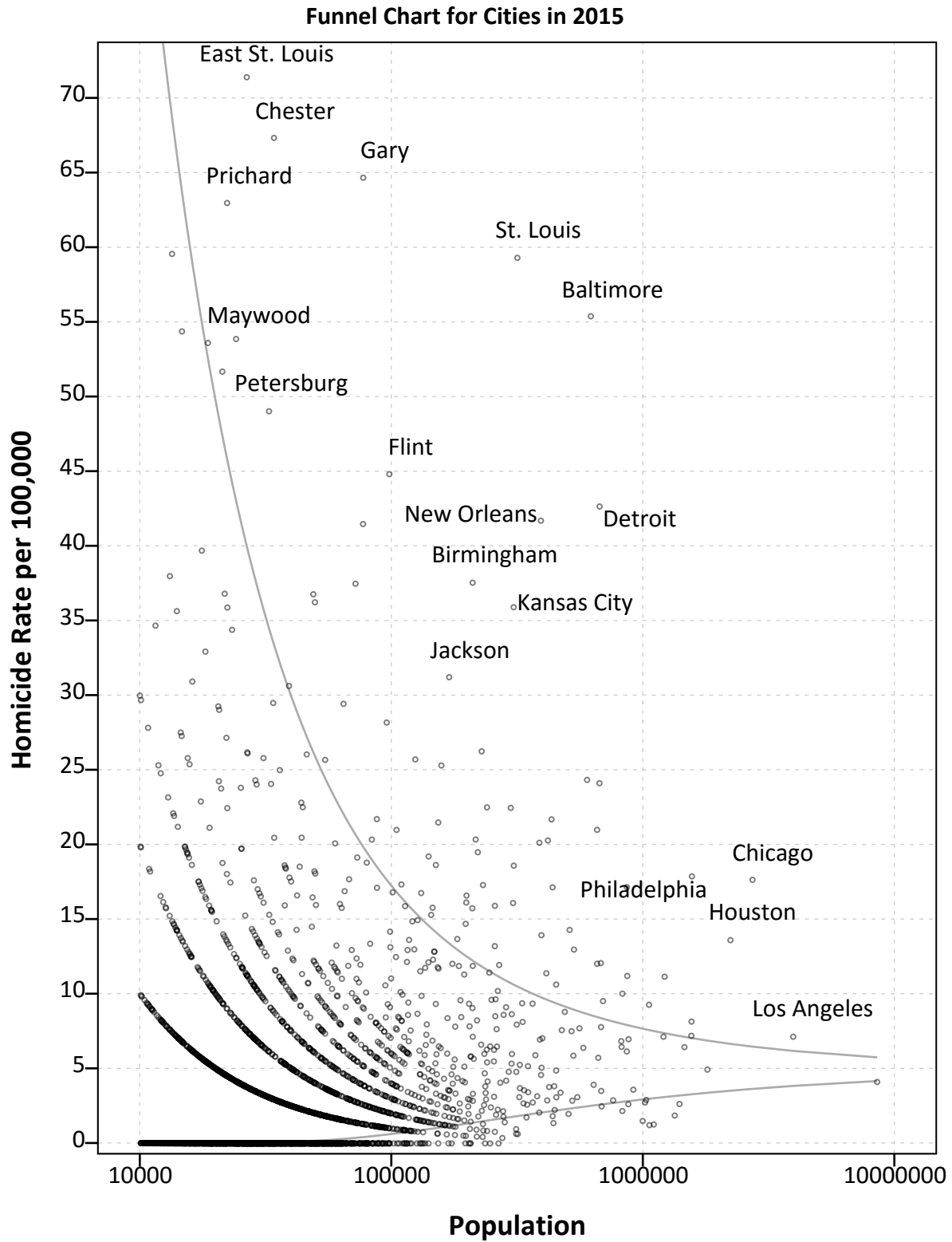


Figure 1: Funnel Chart for Homicide Rates. The grey lines depict the upper and lower bounds of the 99.9% Confidence Interval around the average homicide rate of 4.9 per 100,000.

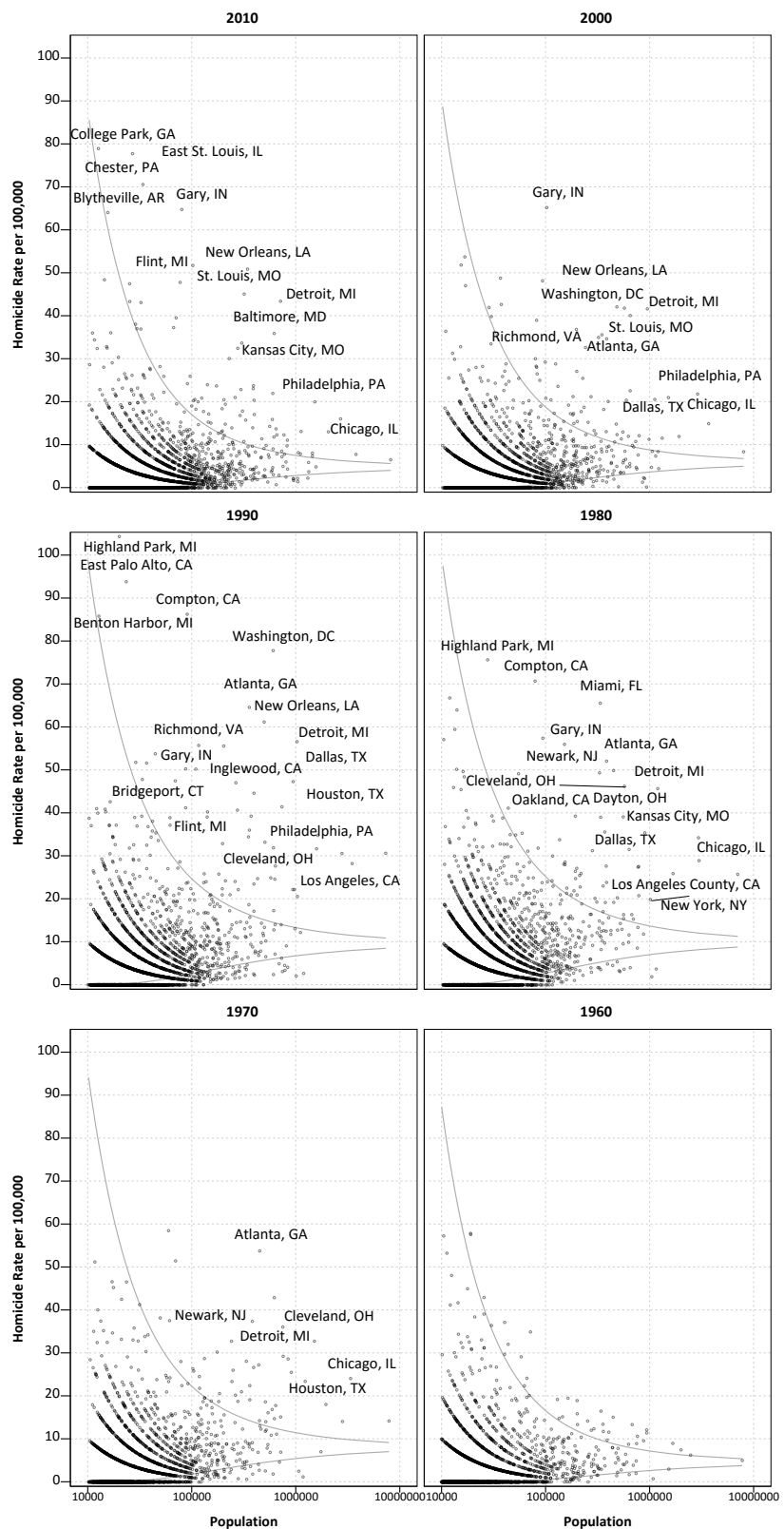


Figure 2: Funnel charts from 2010 to 1960 by decade.

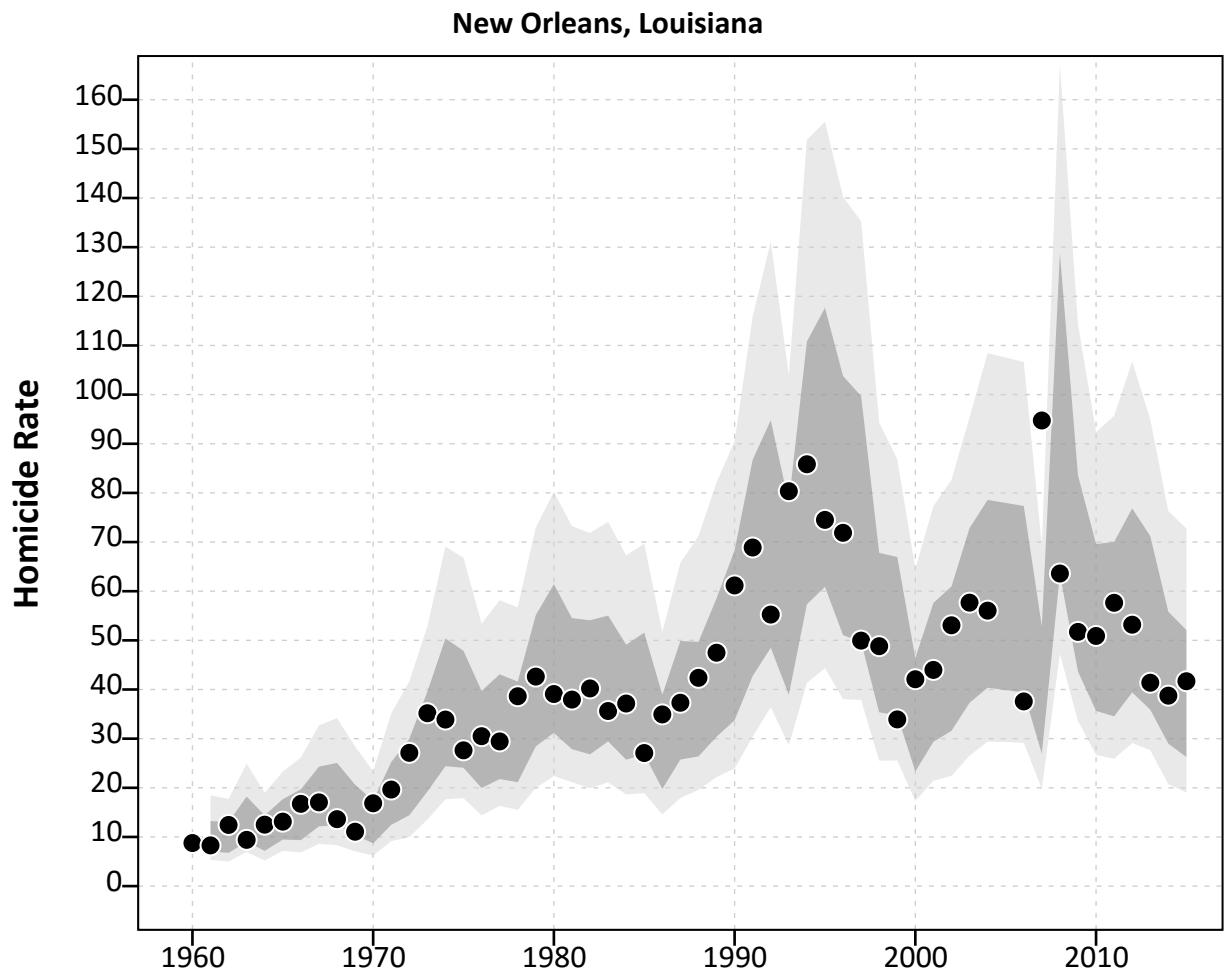


Figure 3: Fan chart for New Orleans homicide rate predictions. The black dot displays the actual homicide rate per 100,000, the dark grey bands denote the 80% prediction interval and the light grey band denotes the 50% prediction interval. The bands are calculated from posterior simulations based on the prior years predictions, and the last data point, 2015, is entirely out of sample.

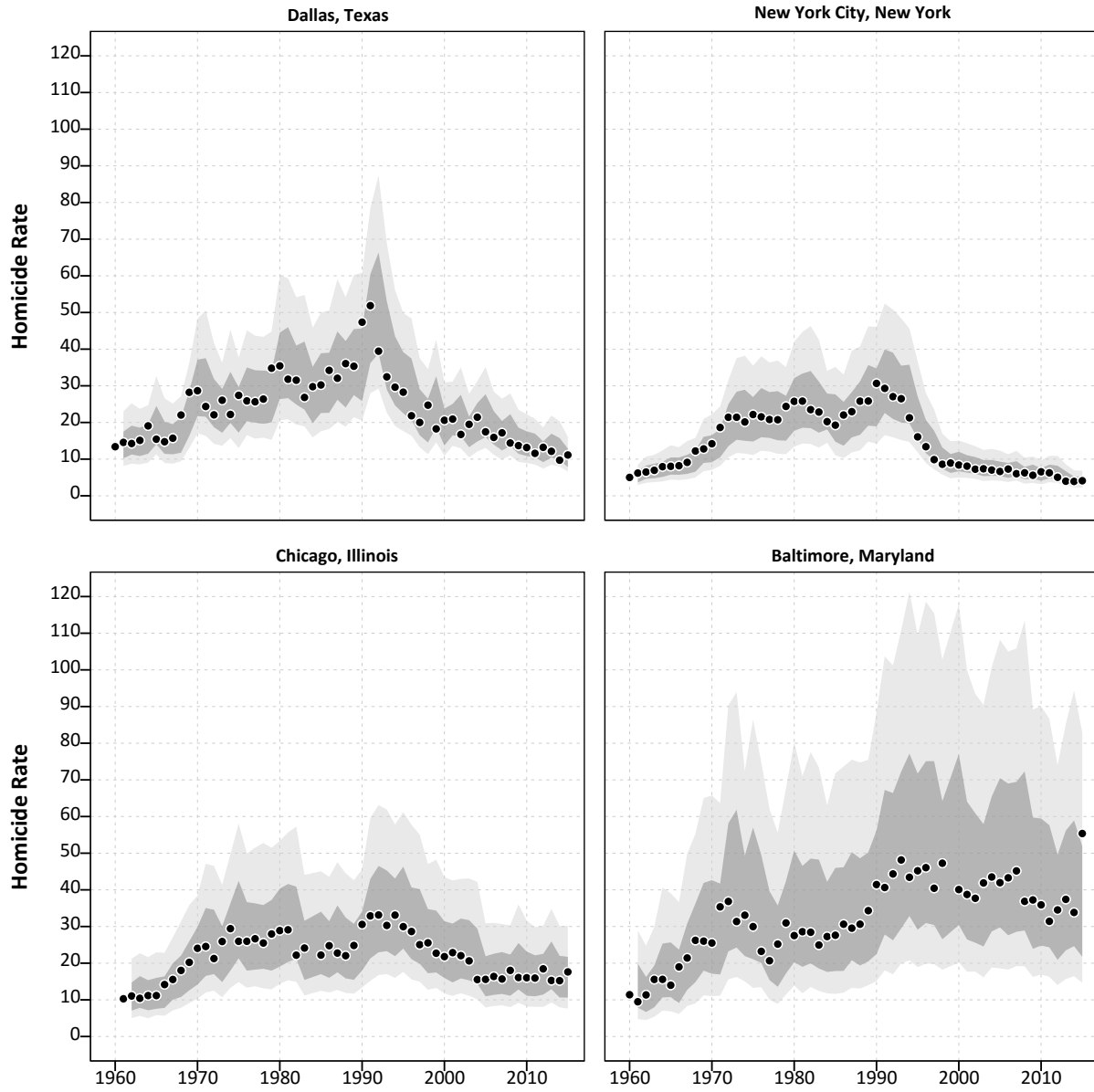


Figure 4: Example fan charts for Dallas, New York City, Chicago, and Baltimore.

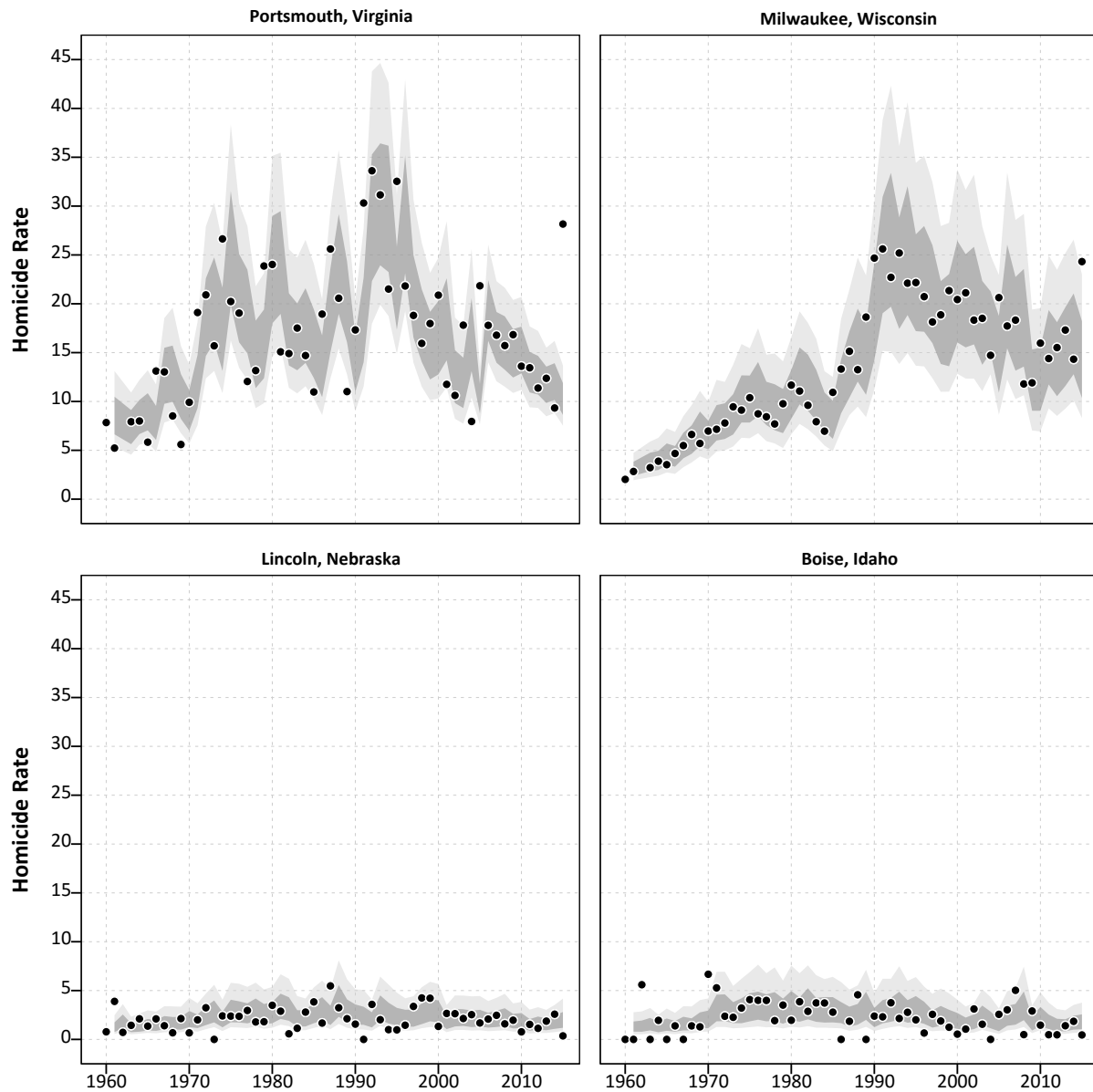


Figure 5: Fan charts for example cities that have more homicides than predicted (Portsmouth and Milwaukee) and fewer homicides than predicted (Lincoln and Boise).

Table 1: Coverage Rates for the 50% and 80% prediction intervals

	Historical Sample		Out of Sample 2015	
	<i>N</i>	%	<i>N</i>	%
<i>Coverage of 50% Intervals</i>				
Low (25th)	7,713	30	150	29
Covered	13,156	51	235	46
High (75th)	5,166	20	129	25
<i>Coverage of 80% Intervals</i>				
Low (10th)	5,066	19	104	20
Covered	18,637	72	358	70
High (90th)	2,332	9	53	10
Total	26,035		515	

Table 2a: 53 Cities that Have Higher Homicide Rates than Predicted in 2015 (based on 80% prediction intervals) (1 to 26)

ORI	Agency Name	Homicides	Population	Hom. Rate	Pred. Low	Pred. High
VA12000	Portsmouth	27	95,877	28.2	7.5	13.7
CA02708	Salinas	40	158,185	25.3	5.7	12.8
FL00600	Broward	7	26,753	26.2	4.7	18.0
TX15512	Waco	22	131,413	16.7	4.6	9.7
SC02604	Horry County	40	231,523	17.3	4.5	11.1
SC00400	Anderson	18	152,973	11.8	5.2	7.1
CA04204	Santa Maria	13	104,355	12.5	1.3	7.8
LA02803	Lafayette	19	127,273	14.9	2.8	11.3
IN04500	Lake	4	47,596	8.4	1.5	5.3
VA11100	Hampton	16	136,381	11.7	4.5	8.9
NH00634	Manchester	6	110,661	5.4	1.2	2.8
VA12300	Roanoke	11	99,827	11.0	3.8	8.5
MI39139	Kalamazoo	5	103,385	4.8	0.5	2.4
IL07207	Peoria	15	116,066	12.9	3.0	10.5
UT01825	West Valley	8	135,744	5.9	1.4	3.6
MI50765	Sterling Heights	5	132,255	3.8	0.4	1.6
CA03015	Orange	7	140,572	5.0	0.7	3.1
MO09500	St. Louis County	28	398,881	7.0	1.2	5.2
NC09203	Cary	5	158,604	3.2	0.6	1.3
TN01901	Nashville Metropolitan	79	658,029	12.0	5.1	10.2
SC00700	Beaufort	11	138,399	8.0	2.6	6.3
WA03900	Yakima	7	86,704	8.1	2.3	6.5
VA08900	Stafford	6	142,286	4.2	1.0	2.6
TX01404	Killeen	17	140,497	12.1	4.7	10.6
TN04701	Knoxville	22	185,638	11.9	5.1	10.4
WV02000	Kanawha	6	91,743	6.5	2.9	5.1

Table 2b: 53 Cities that Have Higher Homicide Rates than Predicted in 2015 (based on 80% prediction intervals) (27 to 53)

ORI	Agency Name	Homicides	Population	Hom. Rate	Pred. Low	Pred. High
WA01700	King	11	271,235	4.1	1.2	2.7
WIMPD00	Milwaukee	146	600,400	24.3	8.3	23.0
NY01455	Cheektowaga Town	2	78,438	2.6	0.5	1.3
CA05002	Modesto	25	210,794	11.9	2.1	10.6
MD02200	Washington	4	101,095	4.0	0.5	2.9
MD01300	Harford	6	211,090	2.8	1.2	1.8
CTCSP00	Connecticut State Police	15	524,558	2.9	0.5	1.8
WA03200	Spokane	5	157,188	3.2	0.8	2.3
OH07700	Summit	2	50,567	4.0	0.5	3.3
ND00902	Fargo	3	118,490	2.5	0.7	1.9
IL08402	Springfield	11	116,875	9.4	1.9	8.8
TX05718	Mesquite	7	145,434	4.8	1.7	4.3
WA01800	Kitsap	6	170,297	3.5	1.5	3.0
MA00311	New Bedford	5	94,833	5.3	1.5	4.8
UT01805	Sandy	2	92,013	2.2	0.7	1.7
NC06700	Onslow	6	112,951	5.3	2.1	4.9
ID00100	Ada	3	114,391	2.6	1.1	2.2
MO04806	Independence	7	117,639	6.0	1.0	5.6
NY03202	Utica	6	61,109	9.8	3.1	9.5
CA0191R	Santa Clarita	7	211,132	3.3	0.5	3.0
WA03103	Everett	5	107,633	4.7	1.4	4.4
WA02700	Pierce	18	393,883	4.6	2.0	4.3
MN08600	Wright	2	109,581	1.8	0.4	1.6
IN00201	Fort Wayne	25	259,712	9.6	2.6	9.5
CO00101	Aurora	25	315,208	7.9	1.7	7.8
UT01800	Salt Lake Cnty Unifd Pd	9	346,894	2.6	1.0	2.5
IL04900	Lake	3	129,449	2.3	0.5	2.3

Table 3a: 104 Cities with Homicide Rates below Predictions (based on 80% prediction intervals) (1 to 35)

ORI	Agency Name	Homicides	Population	Hom. Rate	Pred. Low	Pred. High
MA00926	Lowell	0	110,819	0.0	2.0	7.5
NC09200	Wake	1	193,739	0.5	2.4	6.8
TN03300	Hamilton	1	111,588	0.9	2.8	5.3
AZ01100	Pinal	0	206,311	0.0	1.8	12.1
TX24305	Wichita Falls	1	105,186	1.0	2.8	7.6
TX10115	Pasadena	2	154,986	1.3	3.0	5.5
TN04700	Knox	0	267,044	0.0	1.7	4.1
FL04300	Martin	0	136,503	0.0	1.7	8.2
MI73173	Saginaw	0	40,624	0.0	1.5	6.4
OH02500	Franklin	0	57,988	0.0	1.4	9.0
TX06100	Denton	0	113,789	0.0	1.4	2.4
TX10808	Mcallen	2	140,593	1.4	2.6	4.9
TN07900	Shelby	2	125,039	1.6	2.7	7.2
NV01600	Washoe	1	111,735	0.9	1.9	4.8
VA00203	Albemarle County	0	105,569	0.0	1.0	2.9
CA04106	Daly City	0	107,320	0.0	1.0	4.9
CA04400	Santa Cruz	0	133,970	0.0	1.0	4.7
CA03701	Carlsbad	0	113,972	0.0	1.0	4.6
FL00800	Charlotte	0	153,743	0.0	0.9	4.1
WA00603	Vancouver	2	171,076	1.2	2.1	4.4
TX02000	Brazoria	1	117,448	0.9	1.8	4.6
IL01600	Cook	0	94,987	0.0	0.8	4.6
OH04800	Lucas	0	50,307	0.0	0.8	5.1
MI74174	St. Clair	0	98,281	0.0	0.8	3.3
FL03602	Cape Coral	0	173,844	0.0	0.8	4.3
OH07800	Trumbull	0	27,420	0.0	0.8	3.7
FL01100	Collier	2	318,441	0.6	1.4	6.5
IN02000	Elkhart	0	112,716	0.0	0.8	2.5
MI63163	Oakland	0	21,558	0.0	0.8	2.9
MO01002	Columbia	1	118,911	0.8	1.6	8.1
FL00100	Alachua	1	114,081	0.9	1.6	6.2
CA03607	Ontario	3	170,274	1.8	2.5	9.1
MI25125	Genesee	0	39,968	0.0	0.7	3.5
PA03504	Scranton	0	75,087	0.0	0.7	4.7
CA03008	Fullerton	0	140,771	0.0	0.7	3.8

Table 3b: 104 Cities with Homicide Rates below Predictions (based on 80% prediction intervals) (36 to 70)

ORI	Agency Name	Homicides	Population	Hom. Rate	Pred. Low	Pred. High
CA03105	Roseville	0	131,039	0.0	0.7	3.3
SC02300	Greenville	12	350,423	3.4	4.1	9.9
SC00800	Berkeley	4	116,330	3.4	4.1	9.9
NM00701	Las Cruces	3	102,227	2.9	3.6	6.7
CA00103	Berkeley	1	120,387	0.8	1.5	10.9
TX03101	Brownsville	3	184,941	1.6	2.3	5.4
CO02100	El Paso	1	179,105	0.6	1.2	4.0
NC05100	Johnston	2	134,216	1.5	2.1	6.0
NC09000	Union	1	153,012	0.7	1.3	3.4
PA015SP	Sp: Chester County	0	115,230	0.0	0.6	3.6
VA00701	Arlington County Pd	1	230,906	0.4	1.0	2.1
NJ01205	Edison Township	0	102,281	0.0	0.6	1.4
CA03010	Huntington Beach	0	203,233	0.0	0.6	2.6
FL06000	Sumter	1	111,107	0.9	1.5	7.2
OH08300	Warren	0	91,911	0.0	0.6	2.8
CA03342	Murrieta	0	109,495	0.0	0.6	2.7
UT01806	West Jordan	0	112,687	0.0	0.5	1.5
FL01307	Miami Beach	2	92,641	2.2	2.7	12.9
MI81218	Ann Arbor	0	118,730	0.0	0.5	2.3
NB05501	Lincoln	1	276,585	0.4	0.9	4.2
UT02506	Provo	0	115,294	0.0	0.5	1.4
IL09900	Will	1	102,762	1.0	1.4	7.4
TX17802	Corpus Christi	17	324,326	5.2	5.7	11.0
FL00101	Gainesville	2	129,410	1.6	2.0	8.5
CO03000	Jefferson	1	197,932	0.5	1.0	3.8
GA05800	Forsyth	0	206,539	0.0	0.4	2.0
OH01300	Clermont	0	59,931	0.0	0.4	3.0
FL05500	St. Johns	2	205,441	1.0	1.4	5.9
MI47147	Livingston	0	120,294	0.0	0.4	1.5
MD306SP	Sp: Carroll County	0	126,329	0.0	0.4	2.4
NC02900	Davidson	1	112,768	0.9	1.3	2.9
TX04305	Mckinney	2	163,406	1.2	1.6	4.2
CA01933	Inglewood	7	112,450	6.2	6.6	45.4
MI70170	Ottawa	0	236,062	0.0	0.4	1.7
MI41141	Kent	0	247,192	0.0	0.4	1.6

Table 3c: 104 Cities with Homicide Rates below Predictions (based on 80% prediction intervals) (71 to 104)

ORI	Agency Name	Homicides	Population	Hom. Rate	Pred. Low	Pred. High
IL02214	Naperville	0	98,535	0.0	0.3	1.6
NY03200	Oneida	0	80,851	0.0	0.3	1.1
NY00300	Broome	0	91,705	0.0	0.3	1.0
CA05607	Thousand Oaks	0	129,976	0.0	0.3	1.6
WI06800	Waukesha	0	94,666	0.0	0.3	0.9
NY01400	Erie	0	132,770	0.0	0.3	1.0
OH01842	Parma	0	79,655	0.0	0.3	2.0
LA05200	St. Tammany	3	195,157	1.5	1.8	7.0
FL05900	Seminole	2	218,235	0.9	1.2	6.2
GA06900	Hall	2	143,780	1.4	1.7	8.0
OH05000	Mahoning	0	14,790	0.0	0.3	1.8
FL00500	Brevard	3	231,534	1.3	1.6	7.3
NY04500	Saratoga	0	168,890	0.0	0.3	0.8
ID00101	Boise	1	218,844	0.5	0.7	3.8
NY01300	Dutchess	0	116,140	0.0	0.3	0.8
FL02700	Hernando	2	169,726	1.2	1.4	6.7
NY05500	Ulster	0	43,816	0.0	0.2	1.5
AZ00721	Peoria	1	170,215	0.6	0.8	6.2
TX21204	Tyler	3	102,481	2.9	3.2	6.6
TX22001	Arlington	8	387,565	2.1	2.3	5.0
CA04200	Santa Barbara	2	140,954	1.4	1.6	7.9
NY03500	Orange	0	60,431	0.0	0.2	0.6
AZ00705	Chandler	1	258,875	0.4	0.6	4.4
CA03604	Fontana	3	206,982	1.5	1.6	7.1
CA03900	San Joaquin	5	168,830	3.0	3.1	13.5
FL05602	Port St. Lucie	2	176,364	1.1	1.3	4.8
TX24001	Laredo	8	256,280	3.1	3.3	8.5
TX17000	Montgomery	6	441,523	1.4	1.5	3.7
DE303SP	Sp: Sussex County	3	162,797	1.8	2.0	8.4
LA03200	Livingston	2	117,410	1.7	1.84	7.46
CO01800	Douglas	1	199,939	0.5	0.63	2.31
VA08800	Spotsylvania	2	130,392	1.53	1.62	3.37
MI82343	Dearborn	1	94,962	1.05	1.13	4.75
GA02901	Athens-Clarke County	3	120,858	2.48	2.53	9.42

Appendix

This appendix is provided to describe the data manipulations for the UCR data series before modelling. When compiling the full UCR reported data from 1960 through 2015, this results in a total of 913,335 jurisdiction-years. For data analysis, first, all jurisdiction-years that did not report a full 12 months were eliminated ($n = 261,515$). Of those cases, the majority were listed as reporting zero months ($n = 214,790$). Second, jurisdiction-years reporting zero for population served were also eliminated ($n = 165,515$ in the original database), as well as those with missing values for population served ($n = 3,185$). Zero-population jurisdictions occur regularly for agencies serving larger populations, such as state police agencies (Maltz & Targonski, 2002). Third, cases in which negative homicides were reported were eliminated ($n = 35$). After these operations, the original database of 913,335 jurisdiction-years was reduced to 552,555 cases.

The second stage of correcting data values occurred for records that were clearly anomalous. In 1993, three cities reported homicide totals of over 1,000 which were clearly in error – Newport News, Virginia, Atlantic City, New Jersey and Athens, Georgia. These homicide totals were adjusted to 22, 10 and 13 respectively, which were taken from the online UCR data tool, <https://www.ucrdatatool.gov/>. The ICPSR UCR data series for New York City included the homicides occurring during the terrorist attack of 9/11. Subsequently the homicide counts for New York City during 2001 were changed from 3,452 to 649, which is what the NYPD lists currently at

http://www.nyc.gov/html/nypd/downloads/pdf/analysis_and_planning/seven_major_felony_offenses_2000_2015.pdf.

The last data check concerned false ORI codes (the code used to designate each reporting jurisdiction) or changing ORI codes over time. The first two letters of the ORI correspond to the state reporting, and for the years of 1967 and 1970, Fairfax, Virginia reported their ORI as “SC02901” instead of “VA02901”, although the Agency Name and the State were correct in the original data files (so could subsequently be distinguished from “SC02901” Lancaster, South Carolina).

For changing ORI’s over time, initially the authors noticed that the ORI for the Alaska State Troopers changed from “AKASP00” from 1960 through 2013 to “AKAST01” in 2014. After seeing this error, the authors sorted the final dataset by ORI and year and did a manual check to see if any other agencies also had obvious changes in their ORI. The only other case found was Baltimore County, which was reported as “MD00300” for the years of 1967 through 1970, but was otherwise listed as “MD00301” from 1960 through 2015.

Inevitably, we will have missed some anomalous reporting. In particular, while the funnel charts and fan charts do an excellent job of spotting outlying high data values for homicides, because homicide is rare they will not often highlight zero homicides which are actually missing data (Loftin & McDowall, 2010; Maltz, 2010). So if such jurisdiction-years were mistakenly reported as having the full 12 months, such cases would not be highlighted by our charts, unless the cities were very large (such as over 500,000 population).